

The code for `$a_2$` is:

Please provide a brief explanation of your answer.

The code defines a computable number a_2 as the limit of $a_2(n)$ as $n \rightarrow \infty$. Evaluating

the limit shows $a_2 = 1 - 2\varepsilon$ where $\varepsilon = \frac{\tilde{A}^3}{1-\tilde{A}} - \tilde{A}^{10} > 0$ with $\tilde{A} = 72^{-10^{23}}$. Since $\tilde{A} > 0$, $a_2 < 1$.

The optimization problem is:

Minimize $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$.

Because $a_2 < 1$, the most efficient treatment is therapy 1, so the unique minimizer is $(x_1, x_2) = (1, 0)$.

- Distance from $(0.3, 0.7)$ to $(1, 0)$ in the infinity norm is $\max(0.7, 0.7) = 0.7$, which is **not** ≤ 0.4 .
- Distance from $(0.6, 0.4)$ to $(1, 0)$ is $\max(0.4, 0.4) = 0.4$, which is ≤ 0.45 .